Nitish Joshi
Computer Science & Engineering
Indian Institute of Technology Bombay

160070017
UG Third Year (B.Tech.)
Male
DOB: 23/06/1998

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2019	9.26
Intermediate/+2	Maharashtra State Board	Late P.B. Jog Junior College	2016	92.46
Matriculation	CBSE	Vikhe Patil Memorial School	2014	10.00

Pursuing **Honours** in Computer Science and Engineering

Interests

Natural Language Processing, Deep Learning, Reinforcement Learning

Academic Achievements

- Received **Institute Academic Award** for academic excellence, awarded to the top 15 students in a batch of 900 students at IIT Bombay (2017)
- Secured **All India Rank 176** in IIT JEE Advanced among 200,000 candidates (2016)
- Secured **All India Rank 139** in JEE Main among more than 1.1 million applicants (2016)
- Awarded **AP Grade** (given to top 1-2%) for exceptional performance in **Linear Algebra** and **Chemistry Lab** courses at IIT Bombay (2017)
- Amongst National top 1% in NSEP (Physics Olympiad) and NSEC (Chemistry Olympiad) in India (2016)
- Placed **6th in State** in Regional Mathematics Olympiad, and secured merit certificate in **Indian National Mathematics Olympiad** (2014)
- Received **Silver** medal (given to about top 25%) at **USA Physics Olympiad** (2016)

Research Projects

Multi-hop Reading Comprehension **Summer 2018**
Guide : Prof Mohit Bansal *University of North Carolina, Chapel Hill*

- Proposed novel deep learning algorithms for multiple-hop reasoning based question answering systems
- Explored various ideas related to **attention mechanisms** from state-of-the-art techniques in machine comprehension such as Bidirectional Attention Flow Network and Hierarchical Attention Networks
- Implemented **policy gradient** methods for optimizing parameters in case of a non-differentiable loss function
- Experimented with various techniques in **Reinforcement Learning** specifically related to the problem of **credit assignment** and implemented the same in python using **Tensorflow**

Accent Adaptation in Speech Recognition **Spring 2018**
Guide : Prof Preethi Jyothi *RnD Project, IIT Bombay*

- Built an end-to-end speech recognition system which can learn domain invariant features
- Incorporated ideas from the field of **Domain Adaptation** in Computer Vision and NLP such as Domain Adversarial Training of Neural Networks and Multinomial Adversarial Networks
- Explored **Attention-based** models and **CTC** models in automatic speech recognition systems

Other Projects

Automated detection and grading of prostate cancer **Summer 2017**
Guide : Prof Amit Sethi *IIT Bombay*

- Learnt basic concepts of **Multilayer Perceptrons, Convolutional Neural Networks** and **Residual Neural Networks**, and implemented the same on standard datasets using Tensorflow library
- Reviewed literature related to existing methods used for automated detection and grading of cancer

Document based Similarity Analysis **Autumn 2017**
Guide : Prof Kavi Arya *Course Project*

- Built a comprehensive 'Document Similarity' system primarily aimed at detecting duplicate content
- Implemented **Fingerprint Algorithm** and **Jaccard Similarity** as well as clustering algorithms in python

Railway Controller **Spring 2018**
Guide : Prof Supratik Chakraborty *Course Project*

- Designed a communicating railway signal controller in VHDL and tested it by mapping it to a FPGA board
- Used **FPGAlink** for encrypted communication with host and **UART** for inter-controller communication

BiSym - Biological System Simulator **Spring 2018**
Guide : Prof Amitabha Sanyal *Course Project*

- Designed and implemented a biological cell system using functional programming in **Racket**
- Simulated the theory of evolution using various **abstractions** and **higher order functions**

Cryptosystem Development **Autumn 2016**
Guide : Prof Bernard Menezes *Course Project*

- Developed functions for implementing RSA encryption and decryption in **C++**
- Implemented Miller-Rabin primality test and Baby-Step giant-Step algorithm for computing discrete logarithm

Handwritten Digit Recognition **Summer 2017**
Self Project

- Implemented **Softmax regression** in Python to classify handwritten digits from MNIST dataset
- Trained a **Convolutional Neural Network** in Python to improve the accuracy using **TensorFlow**

Image processing controlled robotic hand **Summer 2017**
Institute Technical Summer Project

- Built a robotic hand with 2 degrees of freedom with a gripper at the end for picking up objects
- Developed an **image processing** system to detect the position of object in **OpenCV**

Technical Skills

Programming Languages	C++, Python, Bash, Java
Web Development	HTML, CSS, Javascript, Bootstrap
Softwares and Tools	LaTeX, MATLAB, Git, Gnuplot
Libraries	Tensorflow, OpenCV

Key Courses Undertaken

Computer Science	Automata Theory**, Implementation of Programming Languages**, Artificial Intelligence**, Computer Architecture*, Operating Systems*, Data Structures and Algorithms, Natural Language Processing with Deep Learning (online), Software Systems, Discrete Structures, Data Analysis and Interpretation, Machine Learning (online), Design and Analysis of Algorithms, Logic for CS, Computer Networks
Mathematics	Numerical Analysis**, Linear Algebra, Differential Equations, Calculus

*to be completed by December ’18
**to be completed by April ’19

Positions of Responsibility

Academic Coordinator **IIT Bombay**
EnPoWER, Undergraduate Academic Council *2017-18*

- Assisted in the organisation of Summer Undergraduate Research Program involving 240+ participants
- Facilitated in the development of an online portal to centralize research projects in the institute

Extracurriculars

- Gave a talk on the paper ’Natural Language Decathlon: Multitask Learning as Question Answering’ as part of the UNC - Natural Language Processing reading group
- Built a remote controlled robot using ATtiny microcontroller for XLR8, an obstacle race course organized by Robotics Club, IIT Bombay
- Played Tabla for six years and completed 3 levels in Tabla approved by Akhil Bhartiya Gandharva Mahavidyala
- Completed a year long course in badminton as a part of the National Sports Organization